

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 4A nr. 46.1

$$\begin{aligned}\int \frac{8(x^3 - 1)}{9x^4 + 10x^2 + 1} dx &= 8 \int \frac{x^3 - 1}{9x^4 + 10x^2 + 1} dx = \frac{8}{9} \int \frac{x^3 - 1}{(x^2 + 1) \left(x^2 + \frac{1}{9}\right)} dx \\&\rightarrow \frac{x^3 - 1}{(x^2 + 1) \left(x^2 + \frac{1}{9}\right)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + \frac{1}{9}} \\&\Rightarrow A(i) + B = \frac{i^3 - 1}{i^2 + \frac{1}{9}} = \frac{9}{8} + \frac{9}{8}i \Rightarrow A = \frac{9}{8}, B = \frac{9}{8} \\&\Rightarrow C\left(\frac{1}{3}i\right) + D = \frac{-\frac{1}{27}i - 1}{-\frac{1}{9} + 1} = -\frac{9}{8} - \frac{1}{24}i \Rightarrow C = -\frac{1}{8}, D = -\frac{9}{8} \\&= \frac{8}{9} \int \frac{\frac{9}{8}x + \frac{9}{8}}{x^2 + 1} dx + \frac{8}{9} \int \frac{-\frac{1}{8}x - \frac{9}{8}}{x^2 + \frac{1}{9}} dx \\&= \frac{1}{9} \int \frac{\frac{9}{2}(2x) + 9}{x^2 + 1} dx + \frac{1}{9} \int \frac{-\frac{1}{2}(2x) - 9}{x^2 + \frac{1}{9}} dx \\&= \frac{1}{2} \ln|x^2 + 1| + \text{Bgtan } x - \frac{1}{18} \ln\left|x^2 + \frac{1}{9}\right| - \frac{1}{\frac{1}{3}} \text{Bgtan}\left(\frac{1}{\frac{1}{3}}x\right) + c \\&= \frac{1}{2} \ln|x^2 + 1| + \text{Bgtan } x - \frac{1}{18} \ln\left|x^2 + \frac{1}{9}\right| - 3 \text{Bgtan}(3x) + c\end{aligned}$$

## References